

Mechanical Design & Simulation Tasks & Instructions — CodeAlpha

Mechanical Design & Simulation — Internship Overview

This internship program provides practical training in mechanical design principles and simulation techniques. CodeAlpha is dedicated to developing industry-ready engineers capable of designing, analyzing and optimizing mechanical systems. The internship equips students with skills in 2D/3D CAD modeling, finite element analysis (FEA) and simulation of real-world engineering problems. Interns will gain hands-on experience in structural analysis, motion simulation and product design. Through guided mentorship and project-based tasks, participants will develop the technical expertise required to contribute effectively to automotive, aerospace, manufacturing, and industrial design sectors.

Internship Perks

- Internship Offer Letter
 - Completion Certificate (QR Verified)
 - Unique ID Certificate
 - Letter of Recommendation (based on performance)
 - Job Opportunities / Placement Support
 - Resume Building Support
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Instructions for Interns

1. Share your internship status on **LinkedIn**, tagging **@CodeAlpha**.
 2. Complete the **assigned projects** within the mentioned time frame.
 3. Upload your complete file to **Google drive**:
`CodeAlpha_ProjectName`
 4. Post a **video explanation** of your project on LinkedIn.
 5. Submit your completed task using the **Submission Form**.
 6. **Complete any 2 or 3 out of the 4 tasks** listed below (from your domain).
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Mechanical Design & Simulation Task List

(Complete any 2 or 3 of the following tasks)

TASK 1: 3D CAD Model Creation

Design a **mechanical component or assembly** (e.g., gear, crankshaft, bearing housing, or connecting rod) using AutoCAD/SolidWorks/Creo.

- Include proper dimensions and material selection.
 Submit the CAD file + rendered images.
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✓ TASK 2: Structural Simulation (FEA)

Perform a **Finite Element Analysis (FEA)** on a simple component (beam, plate or bracket).

- Apply different **loads and boundary conditions**.
 - Analyze **stress distribution, deformation, and factor of safety**.
👉 Submit simulation report with screenshots and observations.
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✓ TASK 3: Motion Simulation

Create a **mechanism model** (e.g., slider-crank, four-bar linkage or gear train).

- Run a **motion simulation** to analyze velocity, acceleration and forces.
 - Present results in graphical/animated format.
👉 Submit simulation file + short report.
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✓ TASK 4: Mini Project – Real-World Mechanical System

Work on a **mini-project** based on your specialization:

- Automotive: Design & simulate a suspension arm.
 - Aerospace: Structural analysis of a wing component.
 - Manufacturing: 3D model of a CNC machine part with stress analysis.
👉 Submit CAD model + simulation results + 2–3 page project report.
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Important Note

Internship Completion Criteria:

To be eligible for the internship certificate, participants must complete a minimum of **two or three tasks**. Submitting only one task will be considered **incomplete** and certificates will **not** be issued in such cases.



Submission Details

A submission form will be shared in your respective **WhatsApp group**. You are required to submit your completed task only through that form. Please follow the instructions mentioned in the form carefully to ensure your submission is accepted.



Contact Information

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